```
PS C:\Users\TINA\Documents\lap_trinh\DevOps\Devops_Midterm\phase1\app\backend> docker build -t khangtrong/notes-backend:v1 .
[+] Building 23.8s (10/10) FINISHED                                                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                        0.1s
 => => transferring dockerfile: 647B                                                                                        0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:21-jdk                                                  3.5s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                                     0.0s
 => [internal] load .dockerignore                                                                                           0.0s
 => => transferring context: 2B                                                                                             0.0s
 => [1/4] FROM docker.io/library/eclipse-temurin:21-jdk@sha256:e8d57b6c5c73d93212a4384f588309b44bc75fd15b38f78e6a6db1b350dc4ef  17.7s
 => => resolve docker.io/library/eclipse-temurin:21-jdk@sha256:e8d57b6c5c73d93212a4384f588309b44bc75fd15b38f78e6a6db1b350dc4ef3  0.0s
 => => sha256:a3307ab29355ce71f400857b35cca7442414151b6d8d767171d67282f986ad07 157.84MB / 157.84MB                         16.1s
 => => sha256:24ebe7eabaf26192ec8e30edade0a6df98ef2590420dfb0ef843456388a54147 158B / 158B                                  1.0s
 => => sha256:df8fd45c97cdc03c294a3813fd5dbdfccd2e3609de14ffa984b6f160afbb7edf 2.28kB / 2.28kB                               1.4s
 => => sha256:a3629ac5b9f4680dc2032439ff2354e73b06aecc2e68f0035a2d7c001c8b4114 29.73MB / 29.73MB                             9.2s
 => => sha256:220e5f46c2556ad3d4e13b9f9820a6a493d998f1bc423515840a27d7a4f41ffe 22.96MB / 22.96MB                             9.2s
 => => extracting sha256:a3629ac5b9f4680dc2032439ff2354e73b06aecc2e68f0035a2d7c001c8b4114                                   0.9s
 => => extracting sha256:220e5f46c2556ad3d4e13b9f9820a6a493d998f1bc423515840a27d7a4f41ffe                                   0.6s
 => => extracting sha256:a3307ab29355ce71f400857b35cca7442414151b6d8d767171d67282f986ad07                                   1.4s
 => => extracting sha256:24ebe7eabaf26192ec8e30edade0a6df98ef2590420dfb0ef843456388a54147                                   0.0s
 => => extracting sha256:df8fd45c97cdc03c294a3813fd5dbdfccd2e3609de14ffa984b6f160afbb7edf                                   0.0s
 => [internal] load build context                                                                                           1.7s
 => => transferring context: 46.80MB                                                                                        1.6s
 => [2/4] WORKDIR /app                                                                                                      0.3s
 => [3/4] COPY target/*.jar app.jar                                                                                         0.1s
 => [4/4] RUN mkdir -p /app/uploads                                                                                         0.3s
 => exporting to image                                                                                                      1.7s
 => => exporting layers                                                                                                     1.3s
 => => exporting manifest sha256:7a273763cf000599f00b16fc0c37ade2c6f212a8c1a289285f356df9c1d617f0                           0.0s
 => => exporting config sha256:0edc59d0aa073235a82b99c5f8eac1b2de92c3428c525074e8bc239a9064a2e5                             0.0s
 => => exporting attestation manifest sha256:a2242952acc5fd440bec9232cc816b7b2e56c4f8278c937cf014c9c9061fa1d2               0.0s
 => => exporting manifest list sha256:f9a45cd59a216ffa2ea59c453a5ca0fb06926ff556fa5814cbc8b3963f0dad4c                      0.0s
 => => naming to docker.io/khangtrong/notes-backend:v1                                                                      0.0s
 => => unpacking to docker.io/khangtrong/notes-backend:v1                                                                   0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/k2gerb2w423kr0fh451w1zr3k
```

Image 1. Build backend image

```
PS C:\Users\TINA\Documents\lap_trinh\DevOps\Devops_Midterm\phase1\app\backend> docker push khangtrong/notes-backend:v1
The push refers to repository [docker.io/khangtrong/notes-backend]
bfa16ba7375f: Pushed
24ebe7eabaf2: Pushed
bb3c6ac5521c: Pushed
220e5f46c255: Pushed
df8fd45c97cd: Pushed
a3629ac5b9f4: Pushed
a6c8bf93b631: Pushed
a144dea7b648: Pushed
a3307ab29355: Pushed
v1: digest: sha256:f9a45cd59a216ffa2ea59c453a5ca0fb06926ff556fa5814cbc8b3963f0dad4c size: 856
```

Image 2. Push backend image to docker hub

```
PS C:\Users\TINA\Documents\lap_trinh\DevOps\Devops_Midterm\phase1\app\frontend> docker build -t khangtrong/notes-frontend:v1 .
[+] Building 317.8s (16/16) FINISHED                                                                         docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                        0.1s
 => => transferring dockerfile: 863B                                                                                        0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                                                             3.5s
 => [internal] load metadata for docker.io/library/node:20-alpine                                                           3.5s
 => [auth] library/node:pull token for registry-1.docker.io                                                                 0.0s
 => [auth] library/nginx:pull token for registry-1.docker.io                                                                0.0s
 => [internal] load .dockerignore                                                                                           0.0s
 => => transferring context: 2B                                                                                             0.0s
 => [build 1/6] FROM docker.io/library/node:20-alpine@sha256:3960ed74dfe320a67bf8da9555b6bade25ebda2b22b6081d2f60fd7d5d430e9c  9.1s
 => => resolve docker.io/library/node:20-alpine@sha256:3960ed74dfe320a67bf8da9555b6bade25ebda2b22b6081d2f60fd7d5d430e9c        0.0s
 => => sha256:3dcec91425079e7b455efc5f2a18d026450c47c9382c41897620afc6b1424e44 1.26MB / 1.26MB                              0.8s
 => => sha256:41b3afaea3b1b1ab04a268431e10dcace7883019a5da7d326aa35dc9713fcbb5 445B / 445B                                  1.0s
 => => sha256:c2b4197efb6ccd7f8b482ae7800f1c9c78c044ea1925878873000 80bcff6b2c9 42.78MB / 42.78MB                            4.7s
 => => sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612 3.86MB / 3.86MB                              1.9s
 => => extracting sha256:1074353eec0db2c1d81d5af2671e56e00cf5738486f5762609ea33d606f88612                                   0.3s
 => => extracting sha256:c2b4197efb6ccd7f8b482ae7800f1c9c78c044ea1925878873000 80bcff6b2c9                                  3.8s
 => => extracting sha256:3dcec91425079e7b455efc5f2a18d026450c47c9382c41897620afc6b1424e44                                   0.2s
 => => extracting sha256:41b3afaea3b1b1ab04a268431e10dcace7883019a5da7d326aa35dc9713fcbb5                                   0.1s
 => [internal] load build context                                                                                           262.4s
 => => transferring context: 313.27MB                                                                                       262.3s
 => [stage-1 1/2] FROM docker.io/library/nginx:alpine@sha256:b0f7830b6bfaa1258f45d94c240ab668ced1b3651c8a222aefe6683447c7bf55  6.9s
 => => resolve docker.io/library/nginx:alpine@sha256:b0f7830b6bfaa1258f45d94c240ab668ced1b3651c8a222aefe6683447c7bf55          0.0s
 => => sha256:e096540205d5d1ab2c2647d716491020d81e3f167a33606665b193d75540857f 20.18MB / 20.18MB                            2.9s
 => => sha256:0abf9e5672665202e79f26f23ef5dbd12558e2ea51ac32807922ab76fdb24ab0 1.40kB / 1.40kB                              1.0s
 => => sha256:085c5e5aaa8eb4b957ecf253c74f16a6a5551231de3fb7c3ac74814a6bf17e06 1.21kB / 1.21kB                              0.7s
 => => sha256:33f95a0f3229b49e777082e801b882b13fcc5b4e389410ce8eb066f4d58c71b9 403B / 403B                                  0.9s
 => => sha256:da7c973d8b92a1555060972c8849a332c93bfe2608c11faeee2098c4cfbe8c3d 953B / 953B                                  0.8s
 => => sha256:567f84da6fbd4287d40a5837485469435c40a81f9a94e98395b6385d3600643a 626B / 626B                                  0.7s
 => => sha256:25f453064fd3e8a9754b6e51b86c637e13203cbfc748fcf73f3c8b2d10816ae3 1.86MB / 1.86MB                              0.9s
 => => extracting sha256:25f453064fd3e8a9754b6e51b86c637e13203cbfc748fcf73f3c8b2d10816ae3                                   0.3s
 => => extracting sha256:567f84da6fbd4287d40a5837485469435c40a81f9a94e98395b6385d3600643a                                   0.1s
 => => extracting sha256:da7c973d8b92a1555060972c8849a332c93bfe2608c11faeee2098c4cfbe8c3d                                   0.0s
 => => extracting sha256:33f95a0f3229b49e777082e801b882b13fcc5b4e389410ce8eb066f4d58c71b9                                   0.1s
 => => extracting sha256:085c5e5aaa8eb4b957ecf253c74f16a6a5551231de3fb7c3ac74814a6bf17e06                                   0.1s
 => => extracting sha256:0abf9e5672665202e79f26f23ef5dbd12558e2ea51ac32807922ab76fdb24ab0                                   0.1s
 => => extracting sha256:e096540205d5d1ab2c2647d716491020d81e3f167a33606665b193d75540857f                                   1.6s
 => [build 2/6] WORKDIR /app                                                                                                0.6s
```

Image 3. Build frontend image

```
PS C:\Users\TINA\Documents\lap_trinh\DevOps\Devops_Midterm\phase1\app\frontend> docker push khangtrong/notes-frontend:v1
The push refers to repository [docker.io/khangtrong/notes-frontend]
0abf9e567266: Waiting
da7c973d8b92: Pushed
7b3eec2d0d05: Pushed
1074353eec0d: Pushed
25f453064fd3: Pushed
567f84da6fbd: Pushed
e096540205d5: Pushed
29943ec2d465: Pushed
33f95a0f3229: Pushed
085c5e5aaa8e: Pushed
v1: digest: sha256:36247f49a388fec8f4f1d7d9b60039b5a52984041686cacf06aed26277617cba size: 856
```

Image 4. Push frontend image to docker hub

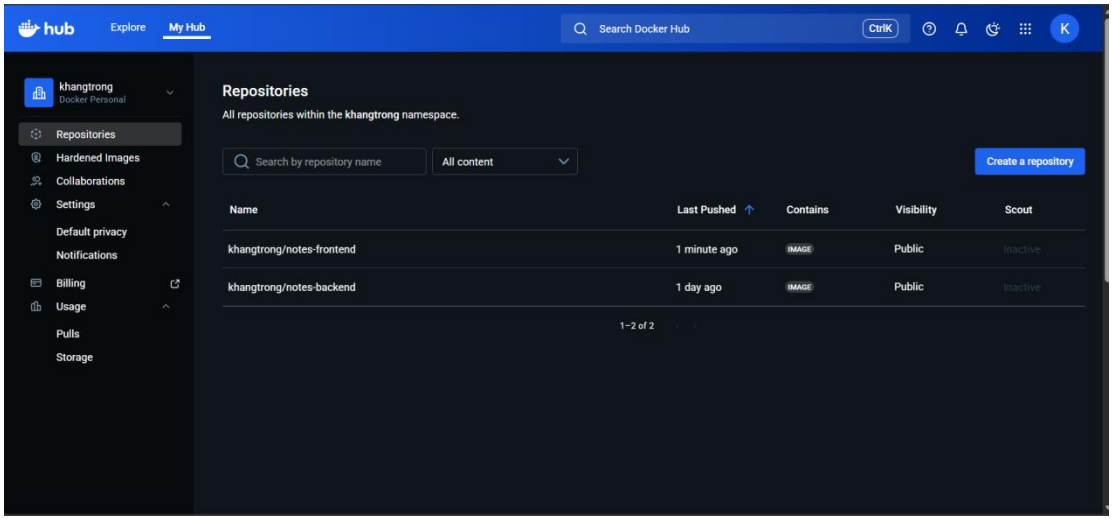


Image 5. List of image on docker hub

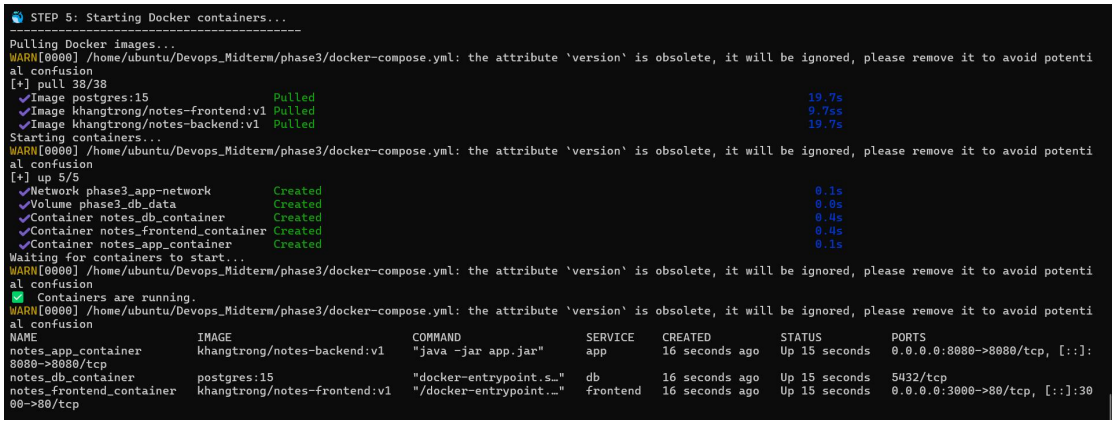


Image 6. Docker containers started using Docker Compose

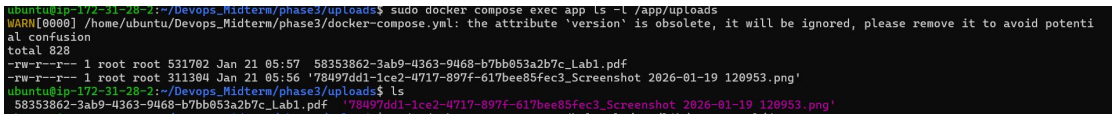
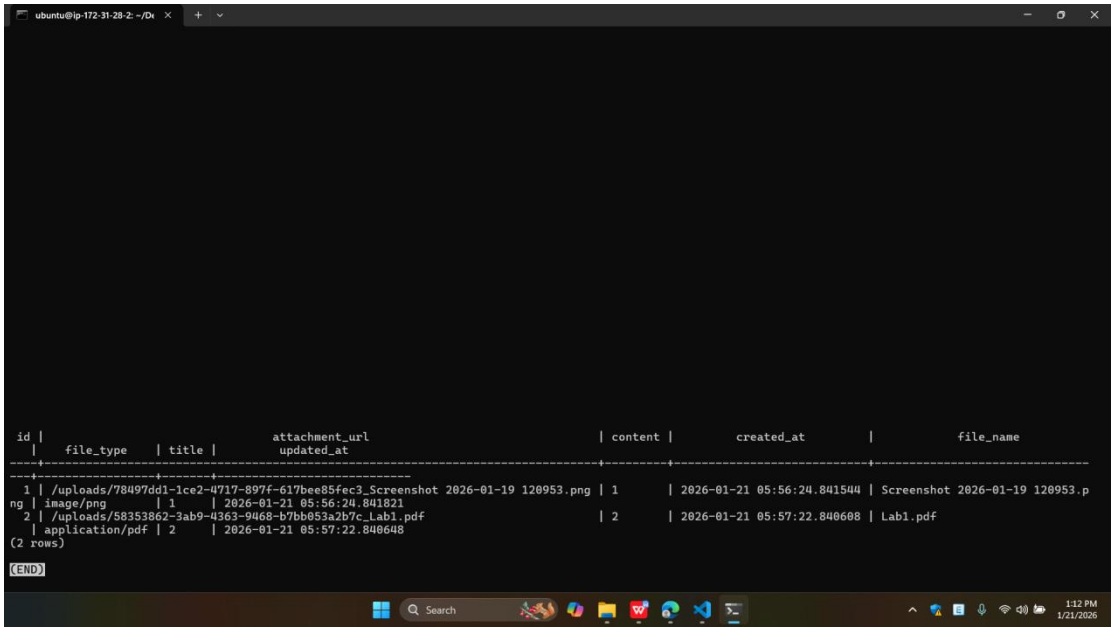


Image 7. Data Persistence via Bind Mount



```
 id |                                attachment_url                                | content |         created_at         |            file_name
   |     file_type     | title |         updated_at
----+------------------------------------------------------------------------------+---------+----------------------------+------------------------------
---+-----------------+-------+----------------------------
  1 | /uploads/78497dd1-1ce2-4717-897f-617bee85fec3_Screenshot 2026-01-19 120953.png | 1       | 2026-01-21 05:56:24.841544 | Screenshot 2026-01-19 120953.p
ng | image/png       | 1     | 2026-01-21 05:56:24.841821
  2 | /uploads/58353862-3ab9-4363-9468-b7bb053a2b7c_Lab1.pdf                        | 2       | 2026-01-21 05:57:22.840608 | Lab1.pdf
   | application/pdf | 2     | 2026-01-21 05:57:22.840648
(2 rows)

(END)
```

Image 8. Application Data Retrieved from PostgreSQL Database Stored in Docker Volume